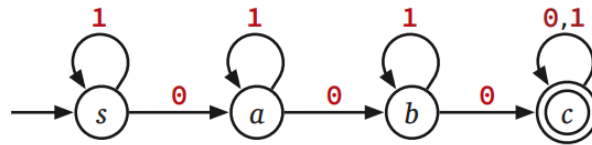


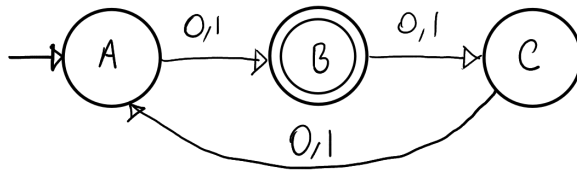
Lecture 3

1. $L_1 = \{w \mid w \in \{0, 1\}^* \text{ all strings containing at least three 0s.}\}$



- s : We've read no 0s.
 - a : We've read one 0.
 - b : We've read two 0s.
 - c : We've read at least three 0s; accept.
2. $L_2 = \{w \mid w \in \{0, 1\}^* \text{ the length of } w \text{ is one more than a multiple of 3}\}$

Multiples of 3: 0, 3, 6, 9,
 lengths: 1, 4, 7, 10, 13,



Lecture 3

3. $L_3 = \{w \mid w \in \{0, 1\}^* \text{ } w \text{ when interpreted as a binary number, is divisible by } 3\}$

Pad 0 to binary string to make it double

Pad 1 to binary string to make it double + 1

Decimal $\div 3 = \text{Binary } \div 3 = \text{remainder}$

$$(0)_{10} \div 3 = (0)_2 \div 3 = 0$$

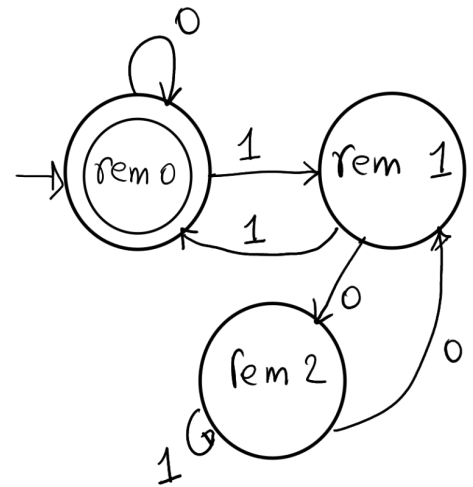
$$(1)_{10} \div 3 = (1)_2 \div 3 = 1$$

$$(2)_{10} \div 3 = (10)_2 \div 3 = 2$$

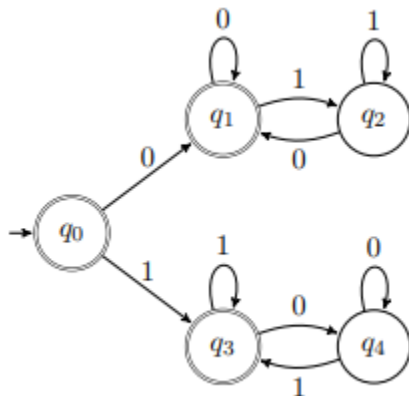
$$(3)_{10} \div 3 = (11)_2 \div 3 = 0$$

$$(4)_{10} \div 3 = (100)_2 \div 3 = 1$$

$$(5)_{10} \div 3 = (101)_2 \div 3 = 2$$

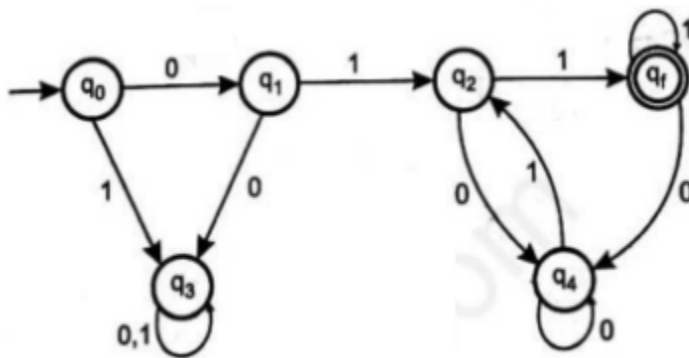


4. $L_4 = \{w \mid w \in \{0, 1\}^* \text{ } w \text{ contains an equal number of occurrences of the substrings } 01 \text{ and } 10\}$



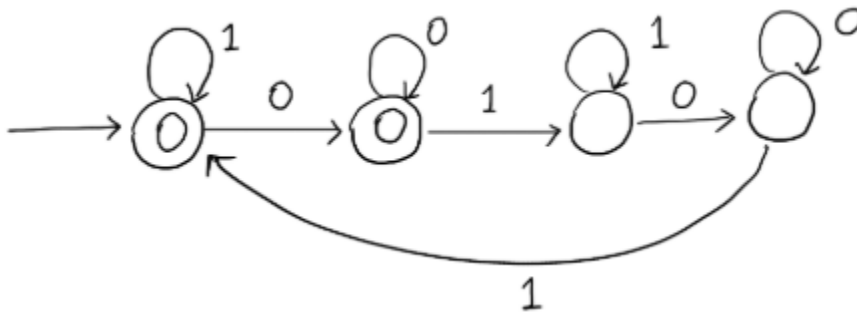
5. $L_5 = \{w \mid w \in \{0, 1\}^* \text{ where the set of strings that begins with } 01 \text{ and ends with } 11\}$.

Lecture 3

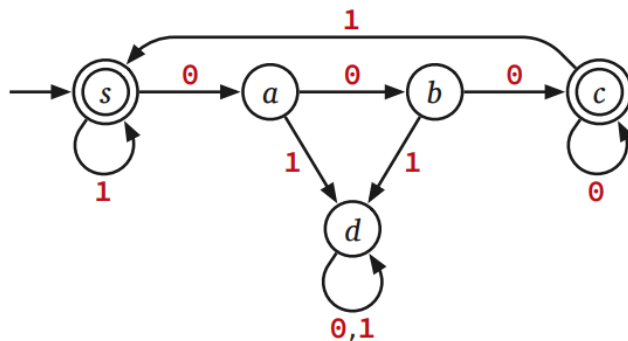


q_3 is the dead state.

6. $L_6 = \{w \mid w \in \{0, 1\}^* \text{ 01 appears even number of times in } w \text{ as a substring} \}$.



7. $L_7 = \{w \mid w \in \{0, 1\}^* \text{ All strings in which every run of 0s has length at least 3} \}$



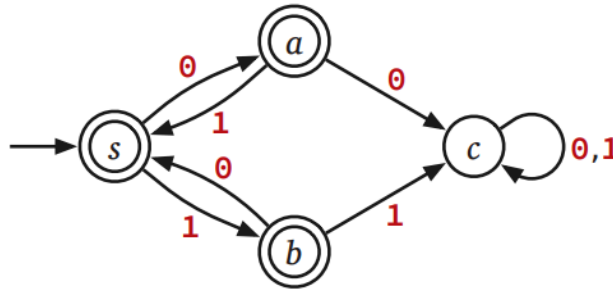
- s : We did not just read a 0
- a : We've read one 0 since the last 1 or the start of the string.
- b : We've read two 0s since the last 1 or the start of the string.
- c : We've read at least three 0s since the last 1 or the start of the string
- d : We've read the substring 01 or 001; reject.

Lecture 3

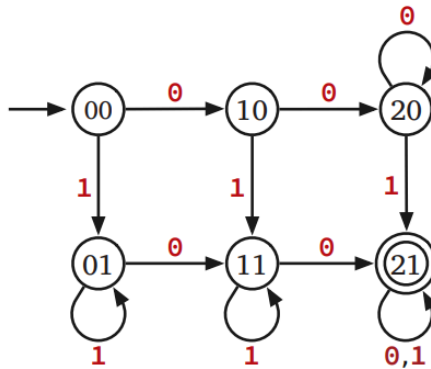
8. $L_8 = \{w \mid w \in \{0, 1\}^* \text{ all strings } w \text{ such that in every prefix of } w, \text{ the number of 0s and 1s differ by at most } 1\}$

L_7 can be 010101, 1010101 (0 and 1 alternate) 0011 not valid.

Solution: This is the same as the set of strings that alternate between 0s and 1s.



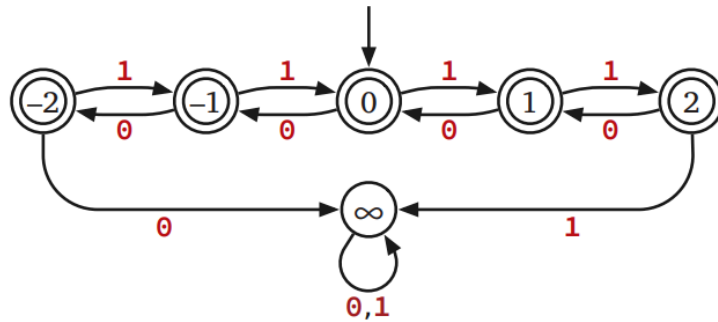
9. $L_9 = \{w \mid w \in \{0, 1\}^* \text{ all strings containing at least two 0s and at least one 1.}\}$



Each state is labeled with a pair of integers. The first integer indicates the number of 0s read so far (up to 2), and the second indicates the number of 1s read so far (up to 1). ■

Lecture 3

10. $L_{10} = \{w \mid w \in \{0, 1\}^* \text{ all strings } w \text{ such that in every prefix of } w, \text{ the number of 0s and 1s differ by at most } 2\}$



The fail state ∞ indicates that we have read some prefix where the number of 0s and 1s differ by more than 2. Each of the other states $-2, -1, 0, 1, 2$ indicates the number of 1s minus the number of 0s of the prefix read so far. ■